

Grade 6 Accelerated
Unit 12
Lesson 2 Homework

Name Answer Key
Date _____
Period _____

Solve each problem using proportional reasoning. Show your work.

1. Sashia, a competitive swimmer, swims at a constant speed of 50 meters in 56 seconds. To the nearest tenth of a second, how long does it take her to swim 228 meters?

$$x = 255.36$$

$$x = 255.34 \text{ seconds}$$

$$\frac{50}{56} = \frac{228}{x}$$

2. To make a shade of paint called international orange, mix 3 quarts of yellow paint with 24 cups of red paint. How much red paint should be mixed with 4.5 cups of yellow paint to make international orange?

$$\frac{12x}{12} = \frac{108}{12}$$

$$x = 9$$

9 cups of red paint are needed

3 quarts = 12 cups

$$\frac{12}{24} = \frac{4.5}{x}$$

3. It takes an ant farm 4 days to consume $\frac{1}{6}$ of an apple. At that rate, in how many days will the ant farm consume 9 apples?

$$x = 216$$

$$\frac{\frac{1}{6}}{4} = \frac{9}{x}$$

In 216 days the ants will consume 9 apples.

4. Raspberries cost \$3.23 per $\frac{1}{2}$ pint. How much would it cost to purchase 3 pints of blackberries?

$$x = 19.38$$

It would cost \$19.38 to purchase 3 pints of raspberries.

$$\frac{0.5}{3.23} = \frac{3}{x}$$

5. Thiago is going on a road trip with his family. He wants to find out how long he will be in the car. His mother told him that their destination is 2,140 miles away and she will be driving at a constant 60 miles per hour. To the nearest tenth of an hour, how long will it take Thiago and his family to reach their destination?

$$x = 35.\bar{6}$$

It will take Thiago and his family 35.7 hours to reach their destination.

$$\frac{60}{1} = \frac{2140}{x}$$

6. An 18-inch pizza from Pizza Pi is cut into 12 slices and costs \$9.95. Rita has enough pizza for each person to eat 2 slices. If Rita has \$29.85 worth of pizza, how many people can she feed?

$$\frac{9.95}{12} = \frac{29.85}{x}$$

$$x = 36$$

Divide by 2 slices per person, so 18 people can eat.

Rita can feed 18 people.

7. Cashews cost \$63.00 per 5 pound bag. How much would a 2 pound bag of cashews cost?

$$\frac{63}{5} = \frac{x}{2}$$

$$126 = 5x$$

$$x = 25.2$$

A 2 pound bag of cashews would cost \$25.20.

8. It takes Richard 30 minutes to mow $\frac{1}{4}$ of an acre of grass. If his yard is 1.6 acres, how many hours will it take him to mow the lawn?

$$\frac{30}{.25} = \frac{x}{1.6}$$

$$48 = .25x$$

$$x = 192 \text{ min.}$$

$$\frac{192}{60} = 3.2$$

It will take Richard 3.2 hours to mow the lawn.

9. Randi wants to paint the rooms in her house. It takes her 24 hours to paint all 4 rooms with 1 coat of paint each. How many rooms can she paint each day if she can spend 8 hours a day painting?

$$\frac{24}{4} = \frac{8}{x}$$

$$24x = 32$$

$$x = 1\frac{1}{3}$$

She can paint $1\frac{1}{3}$ rooms a day.

10. Typing at a rate of 40 words per minute is considered a good speed. If a student has to type a paper that contains 700 words, how many minutes will it take the student to type the paper?

$$\frac{40}{1} = \frac{700}{x}$$

$$40x = 700$$

$$x = 17.5$$

It will take the student 17.5 minutes to type the paper.